Testing the Efficacy of Micro-Blaze Detergent on PAH Removal from Firefighter Uniforms

Erica Babusci[1], Claudia K. Gunsch[1], and Heather Stapleton[2]

1. *Department of Civil and Environmental Engineering, Duke University*
2. *Nicholas School of the Environment, Duke University*

614-600-6861
Erica.babusci@duke.edu

Background

Polycyclic aromatic hydrocarbons (PAHs) are a group of harmful chemicals that have carcinogenic and mutagenic properties.[1,2,4] These compounds are formed through the partial combustion of organic material and accumulate on firefighter PPE, leading to long-term exposure and posing serious health effects.[1] *Micro-Blaze PPE DE-CON Cleaner* (MB) is a product marketed as a spore-containing detergent that not only removes the PAHs but degrades them into nontoxic byproducts.[3]

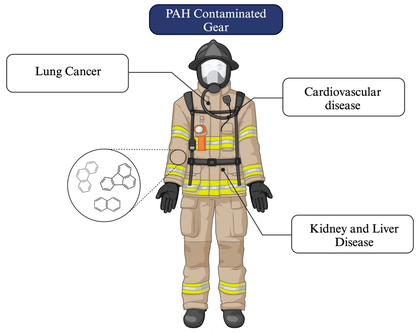


Figure 1: Overview of PAH exposure and effect on PAHs



This project aims to investigate the efficacy of this product for decontaminating firefighter PPE. Using cuttings from a used firefighter jacket and different treatment types, the project investigates the efficacy of using Micro-Blaze DE-CON cleaner to remove and degrade PAHs from the firefighter jacket.

Figure 2: Micro-Blaze PPE DE-CON cleaner

Objectives

Evaluate the effectiveness of Micro-Blaze treatment compared to tap water and standard detergents in removing PAHs from firefighter gear.

Methods

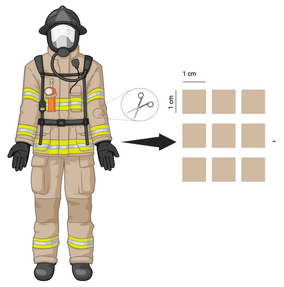


Figure 3: Schematic of firefighter cuttings

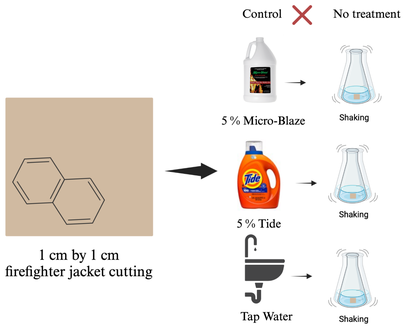


Figure 4: Experimental groups tested for the study

Sample Preparation

A section from the arm of a firefighter jacket was provided from the Stapleton lab. Samples of 1 cm by 1 cm were cut and used for each experimental group in triplicate.

Treatments

Cuttings were exposed to one of the following wash treatments:

- Control (no treatment)
- 5% Micro-Blaze[3]
- 5% Tide
- Tap water

Wash steps involved 10 mL of wash liquid in a 50 mL beaker and left at room temperature on a plate shaker at 100 rpm for 1 hour.

Data Acquisition

Samples were then air-dried and packaged to be returned to the Stapleton lab for PAH extraction and GC-MS analysis.

Results For Micro-Blaze Treatment

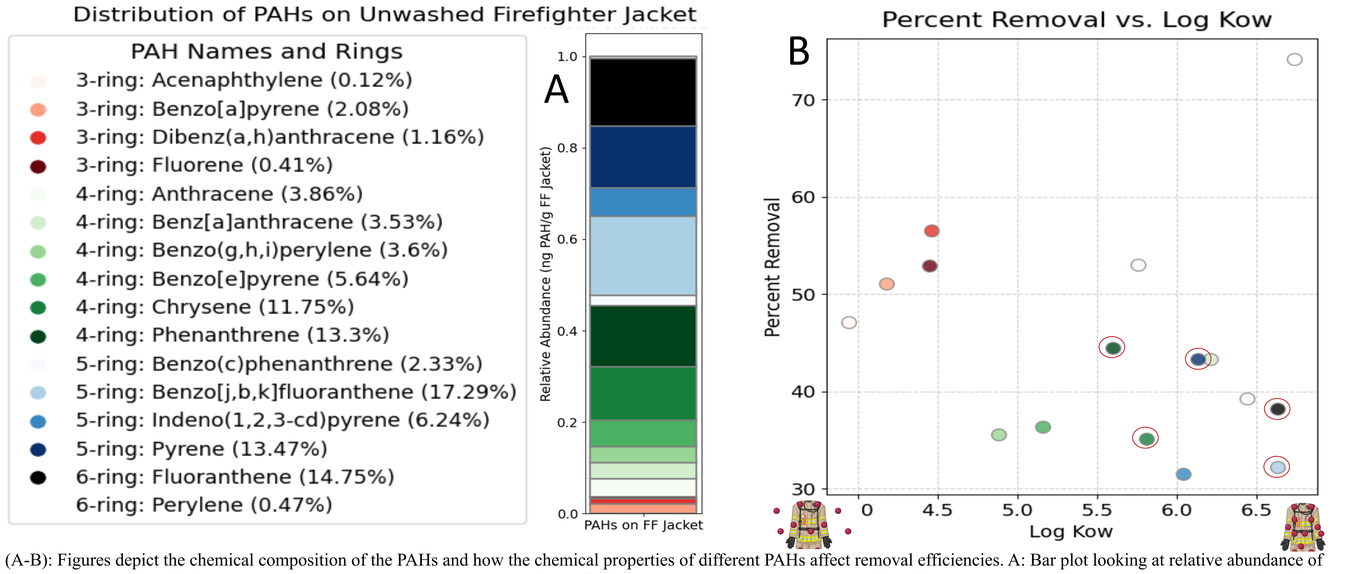


Figure 5 (A-B): Figures depict the chemical composition of the PAHs and how the chemical properties of different PAHs affect removal efficiencies. A: Bar plot looking at relative abundance of each PAH on the control jacket. B: Scatterplot depicting how the percent removal of each PAH is correlated to their respective log Kow. Percent removal is calculated by dividing the Micro-Blaze concentrations to the control/unwashed concentrations

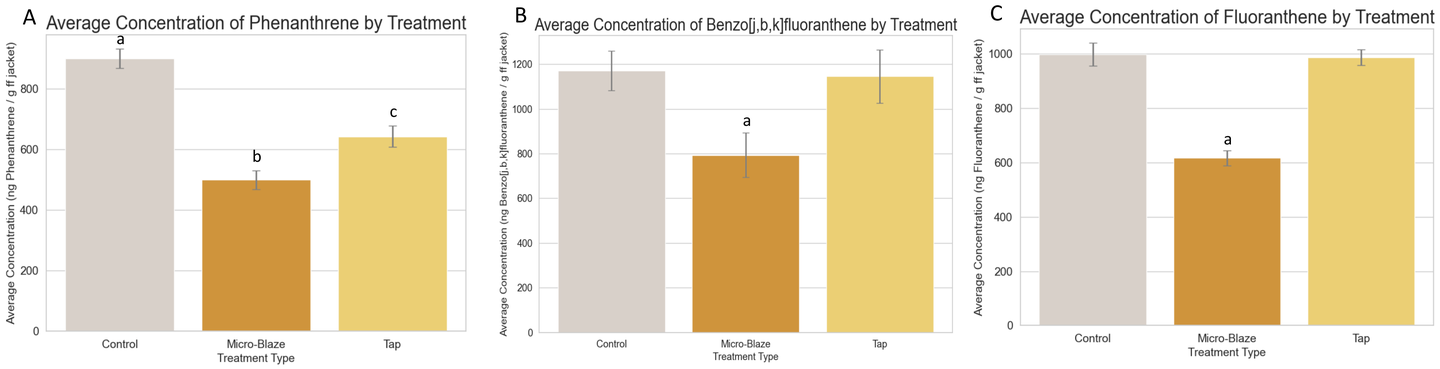


Figure 6 (A-C): Figures are depicting the different in mean PAH concentration between treatment groups. A looks at Phenanthrene (4 rings), B looks at Benzo[j,b,k]fluoranthene (5-rings), and C looks at Fluoranthene (6-rings). Shapiro Wilks test was used to test normality. Fluoranthene was determined to be non-parametric, and a Kruskal-Wallis test was used to test significance. ANOVAs were used for Benzo[j,b,k} fluoranthene and Phenanthrene. Groups with p values < 0.05 were determined to be significant. This trend was seen for 14/16 PAHs

Future Directions

Analyze PAH extractions of the wash water to determine if compounds are being **degraded** or **emulsified**

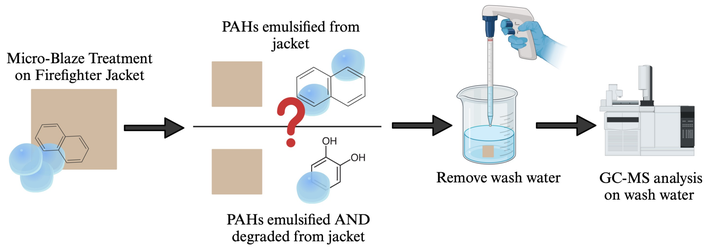


Figure 7: Experimental design for future work exploring PAH removal pathways.

References and Acknowledgements

[1] Kim et al. (2013) *Environment International* [2] Baxter et al. (2014) *J Occup Environ Hyg* [3] Verde Environmental, Inc (2025) *Micro-Blaze Products* [4] Seo, J.-S., Keum, Y.-S., & Li, Q. X. (2009). Bacterial Degradation of Aromatic Compounds.

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